Integrating Sentinel-1 ETAD Into Standard PSI Processing: A Cost-Effective Approach to Phase, Deformation, and Localization Accuracy

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***Abstract*—Persistent scatterer interferometry (PSI) is widely used for surface deformation monitoring, but its accuracy is often limited by atmospheric path delays, orbit errors, and timing inconsistencies. To address these issues, the European Space Agency (ESA) released the Sentinel-1 Extended Timing Annotation Dataset (ETAD), providing refined timing corrections that account for instrument, atmospheric, geophysical, and Doppler (DOP)-related effects. This study introduces PSI+ETAD: a sensor-native PSI processing framework that fully exploits ETAD's range- and azimuth-direction timing annotations to jointly enhance interferometric phase quality, deformation time-series accuracy, and PS 3-D localization precision—thereby delivering simultaneous, cost-effective improvements across all three pillars of operational PSI. Experimental results show that ETAD integration substantially improves phase quality, with reductions of 60.77% in the sum of phase differences (SPDs) and 40.3% in the phase standard deviation (PSD). In deformation velocity estimation, PSI+ETAD achieves a standard deviation (STD) of 5.26 mm/year, compared with 18.64 mm/year for PSI (uncorrected), 2.82 mm/year for PSI+STF (PSI with spatiotemporal filtering), and 6.34 mm/year for PSI+GACOS (PSI with GACOS-based tropospheric (TRO) delay correction), providing a better balance between noise suppression and signal preservation. In time-series analysis, ETAD corrections lower the STD of displacements (SDDs) to 3.65 and the spatiotemporal consistency (STC) to 2.23. Validation against GNSS confirms that PSI+ETAD yields the closest agreement with ground-truth deformation, with the lowest errors (mean absolute error (MAE) of 3.18 mm and root mean square error (RMSE) of 3.91 mm). For geolocation, mean offsets of 5.566 m in range, 1.736 m in azimuth, and 0.974 m in cross-range are corrected, with improvements validated against airborne LiDAR data. These findings demonstrate that integrating ETAD into standard PSI workflows provides a cost-effective and operationally feasible correction strategy, significantly enhancing phase stability, deformation accuracy, and geolocation precision for large-scale InSAR monitoring in urban and tectonically active regions.**

***Index Terms*—Extended Timing Annotation Dataset (ETAD), InSAR, persistent scatterer interferometry (PSI), Sentinel-1.**

I. Introduction

PERSISTENT scatterer interferometry (PSI) utilizes phase observations from a stack of synthetic aperture radar (SAR) images to estimate the deformation history of selected coherent targets, known as persistent scatterers (PSs), over an extended observation period [3], [7], [14]. Benefiting from its all-weather, day-and-night imaging capability and large-scale coverage, PSI has been widely employed in surface deformation monitoring applications.

However, PSI processing is often affected by multiple error sources [12], including atmospheric delays caused by tropospheric (TRO) and ionospheric (ION) variations [25], geodynamic effects such as solid-Earth tides (SETs) and crustal motions [5], [6], [30], bistatic (BIS) azimuth effects [26], [30], Doppler (DOP)-induced range errors, and azimuth biases resulting from frequency modulation mismatches (FMMs).

To mitigate these systematic errors and enhance geolocation and phase fidelity, the European Space Agency (ESA) released the Extended Timing Annotation Dataset (ETAD) for Sentinel-1. ETAD provides high-precision range- and azimuth-timing corrections via refined annotations of internal instrument delays and antenna phase center variations, together with dedicated layers for propagation-path delays (TRO and ION), solid-Earth tidal displacements, DOP-induced range shifts, and azimuthal biases due to FM-rate mismatch.

In other words, integrating ETAD corrections into a standard Sentinel-1 PSI workflow directly strengthens interferogram quality, deformation accuracy, and geolocation precision—three long-standing objectives of the InSAR community [10], [36]. Historically, advances in PSI accuracy have mainly relied on mitigating atmospheric delays, using external atmospheric models [31], spatiotemporal filtering approaches [7], [14], or empirical scene-dependent adjustments [18], [19], [20], [32]. In contrast, ETAD provides a sensor-native and systematic correction framework that extends beyond atmospheric effects by embedding refined annotations of Earth dynamics (e.g., SETs) and instrument-related parameters (e.g., internal delays and antenna phase center variations) directly into the SAR data stream.

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Previous studies have demonstrated that the ETAD product can be used to improve both the geometric accuracy and interferometric phase correction of Sentinel-1 SAR data. Gisinger et al. [10] described the development of ETAD and emphasized its application in correcting timing errors caused by meteorological effects and SAR imaging distortions, achieving geolocation errors below 0.2 m. Sanchez et al. [24] investigated ETAD's capabilities and limitations in interferometric applications, showing that its timing correction data can enhance geometric accuracy and support the generation of interferometric phase screens based on repeat observations. ETAD has also been successfully applied in snow and ice motion monitoring and large-scale interferogram phase correction. Zinno et al. [36] further analyzed ETAD's use in Sentinel-1 DInSAR applications, proposing the recomputation of ETAD atmospheric layers to effectively reduce atmospheric phase contributions and comparing the results with ERA5 data, confirming its effectiveness for medium-resolution DInSAR interferograms and deformation time-series analysis. Collectively, these studies indicate the effectiveness of ETAD in both geolocation and DInSAR. Collectively, these studies validate ETAD's utility in geolocation refinement, single-interferogram phase correction, and DInSAR-based deformation mapping—but none have embedded ETAD into a full PSI processing chain, nor evaluated its impact on the joint optimization of phase stability, deformation time-series fidelity, and PS geolocation precision.

To bridge this gap, we propose PSI+ETAD: the first end-to-end PSI framework that systematically integrates ETAD corrections. Specifically: 1) range-direction timing errors (TRO, ION, SET, and DOP-related) are converted into interferogram-level phase offsets to suppress noise before unwrapping; and 2) azimuth-direction residuals (e.g., FMM) are translated into burst-timing and DOP-centroid refinements to correct PS geolocation drift—all without requiring GNSS or external ground control. This unified strategy treats phase and geometry not as decoupled postprocessing steps, but as physically coupled outcomes of sensor-level timing fidelity.

To rigorously quantify the impact of ETAD integration across all PSI performance dimensions, we conduct a four-way comparative experiment, evaluating: 1) an uncorrected PSI solution; 2) a PSI solution incorporating GACOS-based atmospheric corrections; 3) a PSI solution enhanced by spatiotemporal filtering; and 4) PSI+ETAD—our proposed framework. Validation employs a dual independent ground-truth framework: GNSS time series serve as the reference for deformation time-series accuracy; airborne LiDAR point clouds provide the high-resolution spatial reference for PS geolocation precision. These validations confirm that PSI+ETAD is not an incremental add-on, but a sensor-native, operationally scalable upgrade to standard PSI processing—enabling more reliable large-scale deformation monitoring without external infrastructure.

This article is organized as follows. Section II introduces the Sentinel-1 ETAD, explaining the physical meaning, data sources, and correction logic of its six integrated layers. Section III presents the proposed ETAD-integrated PSI strategy, covering both phase corrections and timing refinements in azimuth and range, and describes how geolocation and phase errors are addressed, along with the evaluation methodology. Section IV outlines the experimental setup, including study area, data sources, processing steps, and parameter settings. Section V presents the experimental results, comparing the proposed PSI+ETAD approach against three reference variants: 1) PSI (uncorrected); 2) PSI+STF (with spatiotemporal filtering); and 3) PSI+GACOS (with GACOS-based atmospheric correction). The evaluation covers: interferogram quality, deformation velocity estimation, time-series stability, and PS geolocation accuracy. Section VI summarizes the main findings and outlines recommendations for future applications of ETAD-integrated PSI processing.

II. ETAD Product: Structure and Key Features

The ETAD is a correction product released by the ESA in 2021, designed to improve the geolocation accuracy of Sentinel-1 SAR data [9], [36]. ETAD provides high-precision timing information by modeling and compensating for various system and geophysical errors, allowing users to achieve accurate geolocation directly from standard Sentinel-1 SAR products.

The ETAD dataset is delivered in the Standard Archive Format for Europe (SAFE). Each ETAD product includes a precise orbit segment specific to the associated Sentinel-1 acquisition, a quick-look processing summary, and two core components: an ETAD network common data format (NetCDF) file, and an ETAD extensible markup language (XML) file, as illustrated in Fig. 1.

This product contains six correction layers. Here, we provide a detailed analysis of each individual correction component.

1) *TRO delay* refers to the signal propagation delay introduced by variations in atmospheric conditions within the troposphere—the lowest layer of the Earth's atmosphere. Key influencing factors include temperature, pressure, and especially water vapor content, which cause spatial and temporal variability in the atmospheric refractive index. These variations distort the travel time of the radar signal, thereby introducing phase delays in the phase measurement.
   The TRO delay $\Delta\tau_{\text{tro}}$ is determined by the length of the signal propagation path and the atmospheric refractive index distribution along the path. The resulting TRO delay can be expressed as

$$\Delta\tau_{\text{tro}} = \frac{2 \cdot \Delta R_{\text{tro}}}{c} \tag{1}$$

   where c denotes the speed of the signal in a vacuum, ΔR_{tro} represents the TRO slant-range path delay at the time of radar data acquisition. It is computed as a weighted summation of the TRO slant-path delays derived from numerical weather model (NWM) outputs at multiple time instants. TRO delay can reach between 2 and 4 m, depending on variations in topography and radar viewing geometry.
   The data of this correction layer were derived from the 4-D atmospheric reanalysis products—spanning both

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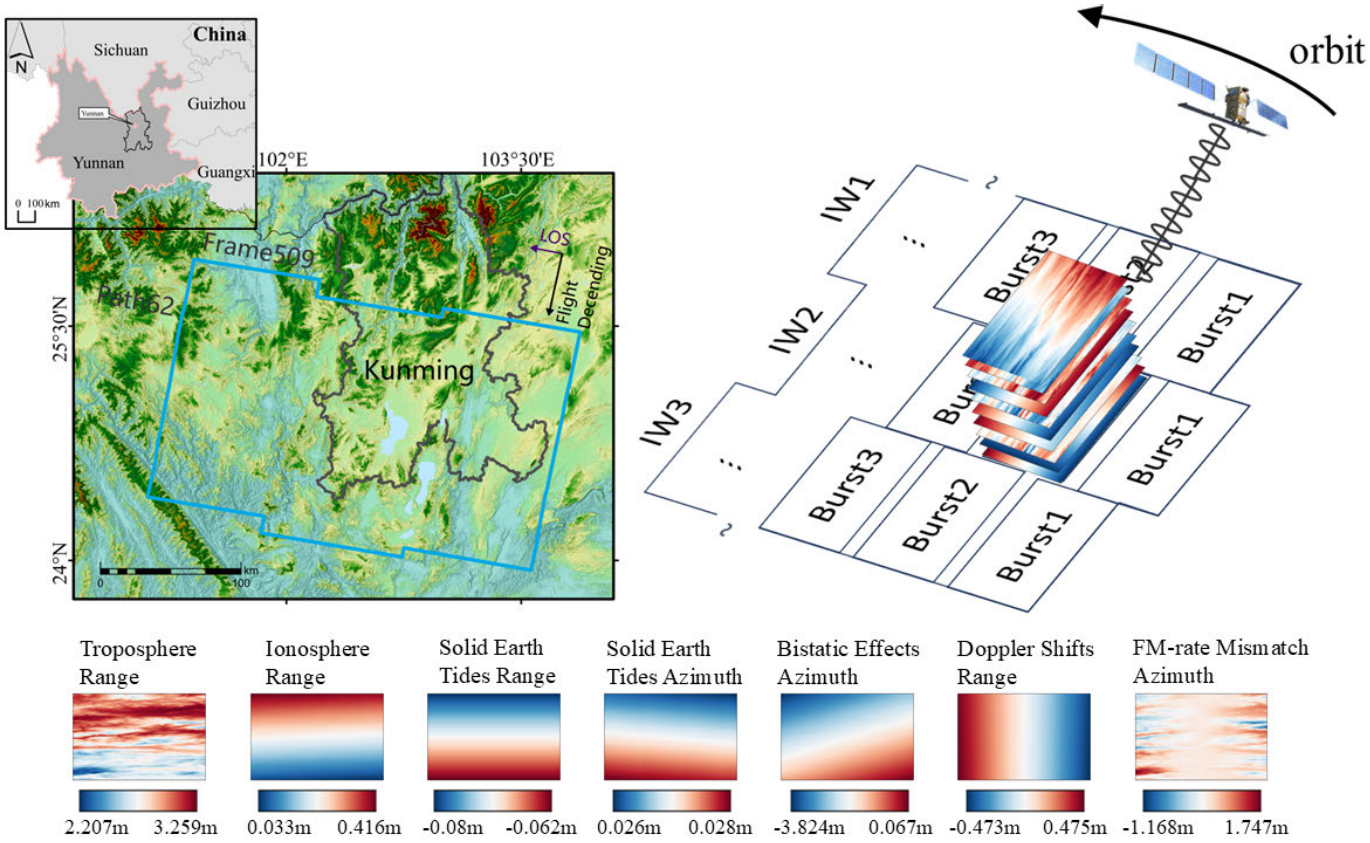


Fig. 1. Sentinel-1 ETAD product structure and correction layer visualization. This figure illustrates the structural composition of the ETAD, and presents the spatial distribution and numerical range of each of its six correction layers using the Kunming study area as an example.

spatial and temporal dimensions—provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). The refractivity distribution along the line of sight between the SAR sensor and the ground target is obtained by interpolating the atmospheric parameter data with discrete time and spatial resolution provided by ECMWF [23]. The resulting refractivity is then integrated along the line of sight to calculate the TRO delay, thereby generating a TRO correction layer [21].

2) *ION delay* refers to the propagation delay experienced by radar signals as they pass through the ionosphere, a dispersive layer of the atmosphere composed primarily of free electrons and ions. Due to its frequency-dependent refractive properties, the ionosphere alters the phase velocity of electromagnetic waves, causing them to travel more slowly along the radar line of sight. This results in a measurable delay in the signal arrival time, particularly affecting L-band and lower frequency SAR systems.

The ION delay $\Delta\tau_{\text{iro}}$ is typically quantified using the total electron content (TEC), which represents the integrated number of electrons along the signal path per unit area. The delay can be approximated by the following equation:

$$\Delta\tau_{\text{iro}} = \frac{2 \cdot C_1}{c \cdot f^2} \cdot E_{TEC}\left(\phi_{TEC}, \lambda_{TEC}, t_g\right) \cdot M\left(z'\right) \cdot \alpha \quad (2)$$

where C_1 is the TEC proportionality constant, set to, set to $40.3 \cdot 10^{16}[\text{m}^3\text{s}^{-2}]$, and f is the radar signal frequency in Hz, $E_{TEC}(\phi_{TEC}, \lambda_{TEC}, t_g)$ denotes the TEC value extracted from the daily global TEC map product at zero-DOP azimuth time t_g, corresponding to the correction grid location defined by the geocentric latitude ϕ_{TEC} and longitude λ_{TEC}. The ION mapping $M(z')$ function depends on the zenith angle at the ION pierce point (IPP), and the scaling factor α, set to 0.9, accounts for the conversion from vertical TEC (vTEC) to slant TEC based on the Sentinel-1 orbit altitude [13]. This inverse square relationship highlights the significant impact of lower frequency signals. For Sentinel-1 operating in the C-band, the ION delay in slant range can be as large as 0.5 m.

The ION delay correction layer is derived from global TEC maps obtained through integration of GNSS observations [27]. These maps provide temporally and spatially resolved electron density distributions, which are projected along the radar line of sight to estimate ION-induced phase delays [22].

3) *SET* delay refers to the propagation delay in radar signals caused by the elastic deformation of the Earth's crust in response to periodic gravitational forces exerted primarily by the Moon and the Sun. These tidal forces induce subtle but measurable vertical and horizontal displacements of the Earth's surface—typically on the order of a few centimeters—which alter the effective distance between the radar sensor and the ground target [5], [6], [30]. As a result, the signal propagation path is slightly lengthened or shortened, introducing small

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delays that can affect the accuracy of interferometric phase measurements, particularly in high-precision InSAR geodetic applications.

The displacement vector $\Delta\mathbf{X}_{SET}$ represents the total deformation due to SETs. It includes contributions from degree-2 to degree-3 radial tides, diurnal and semidiurnal transverse displacements, out-of-phase components, and frequency-dependent corrections. In general, vertical displacements induced by SETs generally range between −25 and + 25 cm, while horizontal shifts may reach up to 6 cm. This 3-D displacement is then mapped to corrections in the azimuth timing and slant-range delay via the zero-DOP geometry.

The corrected azimuth time t_{SET} is determined by solving the zero-DOP condition using Newton iteration

$$\frac{\dot{\mathbf{X}}_s(t_{\mathrm{SET}})\cdot\left(\mathbf{X}_g+\Delta\mathbf{X}_{\mathrm{SET}}-\mathbf{X}_s(t_{\mathrm{SET}})\right)}{\left|\dot{\mathbf{X}}_s(t_{\mathrm{SET}})\right|\cdot\left|\mathbf{X}_g+\Delta\mathbf{X}_{\mathrm{SET}}-\mathbf{X}_s(t_{\mathrm{SET}})\right|}=0. \tag{3}$$

The resulting azimuth correction is

$$\Delta t_{\mathrm{SET}}=t_{\mathrm{SET}}-t_g. \tag{4}$$

Here, t_g is the nominal azimuth time, $\dot{\mathbf{X}}_s$ is the satellite velocity vector, $\mathbf{X}_s(t)$ is the satellite position at time t, $\mathbf{X}_g$ is the grid point position. The corrected two-way slant-range propagation time is computed as

$$\tau_{\mathrm{SET}}=\frac{2\cdot\left|\mathbf{X}_g+\Delta\mathbf{X}_{\mathrm{SET}}-\mathbf{X}_s(t_{\mathrm{SET}})\right|}{c}. \tag{5}$$

Thus, the slant-range delay correction becomes

$$\Delta\tau_{\mathrm{SET}}=\tau_{\mathrm{SET}}-\tau_g \tag{6}$$

where τ_g is the slant-range time.

SET corrections are derived from established geophysical models, such as IERS conventions [2], [5], [6], [8], and tied to a global geodetic reference frame. These models simulate crustal deformations induced by lunar and solar gravitational forces, which are then projected onto radar range and azimuth directions to generate timing corrections applicable to SAR measurements.

4) *BIS azimuth delay* refers to the geometric delay introduced in radar measurements due to the asymmetry between the transmitter and receiver positions in BIS or quasi-BIS SAR configurations. In such cases, the signal propagation paths to and from the target are not identical, resulting in unequal travel times. This path asymmetry introduces systematic biases—particularly in the azimuth direction—that can affect the accurate geolocation of targets in SAR image formation.

The BIS azimuth delay correction Δt_{BIS} at a specific slant-range time τ_g is computed using the mid-swath time τ_{mid}, the pulse repetition interval (*PRI*), and the number of traveling pulses applicable to the currently processed burst (*rank*)

$$\Delta t_{BIS}=\frac{\tau_{mid}}{2}+\frac{\tau_g}{2}-rank\cdot PRI. \tag{7}$$

This results in subpixel distortions and range-dependent azimuth shifts of up to 4 m in Sentinel-1 IW products [26], [30].

The BIS azimuth residuals are estimated using the ancillary annotation data embedded in the Sentinel-1 single look complex products. These residuals reflect variations in satellite positioning and acquisition geometry, and are used to compute and apply azimuth timing corrections to mitigate BIS effects in SAR image formation [17].

5) *DOP-induced range shifts* arises from the relative motion between the radar platform and the target, resulting in a frequency shift in the received signal. In SAR systems, this frequency shift directly affects the focusing and geometric positioning of the radar echoes, particularly in the range direction. If uncorrected, it can introduce timing errors and degrade the accuracy of SAR image formation and geolocation.

To correct for this, the DOP-centroid frequency $f_{DC,g}$ at the grid location is computed, followed by the correction term $\Delta\tau_{dop}$ in slant-range delay

$$\Delta\tau_{dop}=\frac{f_{DC,g}}{K_r} \tag{8}$$

where K_r is the chirp FM-rate. The magnitude of the DOP frequency shift is proportional to the relative motion between the sensor. This effect can introduce range offsets of approximately ±0.4 m.

This correction layer is derived from the ancillary annotation data of Sentinel-1 SLC products. It accounts for the range offset caused by DOP-centroid variations in TOPS mode [1], where the DOP shift influences the focusing geometry and introduces systematic distortions that require compensation during image processing [11].

6) *FMM* arises when the annotated azimuth FM-rate deviates from the true FM-rate derived from geometry. This mismatch affects the time-domain focusing of the azimuth signal, leading to temporal shifts and frequency response distortions [4]. As a consequence, it results in systematic azimuthal displacements in the focused SAR image, potentially causing geolocation errors, target misregistration, and degraded image quality.

The FMM correction is evaluated by comparing the annotated FM-rate $\kappa_{a,b}$ with geometrically described value

$$\Delta t_{FMM}=f_{DC,g}\cdot\left(\frac{1}{-\kappa_{a,b}}-\frac{1}{-\kappa_{a,true}}\right). \tag{9}$$

This relationship highlights the sensitivity of azimuth positioning to even small FM-rate inconsistencies. The true FM-rate induced by topography can cause azimuthal displacements of up to 1 m.

The azimuth offset correction due to FM-rate mismatch is calculated using the Sentinel-1 instrument processing facility (IPF), which incorporates high-resolution topographic information from the Copernicus 90-m DEM along with annotation data from Sentinel-1 SLC products. These corrections are terrain-dependent and are computed under assumed topographic conditions to mitigate azimuthal misalignment effects in focused SAR imagery [9].

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III. Methodology

A. Operational Challenges in ETAD Integration for PSI

In this study, we present a systematic strategy to incorporate the Sentinel-1 ETAD into both the interferometric phase refinement and PS points geolocation process. The correction process operates on two complementary levels, as illustrated in Fig. 12.

The temporal resolution of the ETAD dataset is consistent with that of standard Sentinel-1 SAR data, with a revisit interval of 12 days. In terms of spatial resolution, the azimuth resolution is approximately 200 m. The range resolution varies depending on the local incidence angle and terrain conditions, but it typically corresponds to about 120 m in the slant-range direction, which translates to approximately 160–250 m in ground range.

To achieve spatial consistency between the original ETAD grid and the target reference grid, a bilinear interpolation-based resampling approach was adopted. Let (x^E, y^E) denote the coordinates of the original ETAD grid, and (x', y') denote the coordinates of the target reference grid. The ETAD parameter field is defined as $f(x_i^E, y_j^E)$. For any target grid point (x', y'), the bilinear interpolation function is expressed as

$$f_E\left(x', y'\right) = a_0 + a_1 x_i^E + a_2 y_j^E + a_3 x_i^E y_j^E. \tag{10}$$

The coefficients a_0, a_1, a_2, and a_3 are determined using the four surrounding ETAD grid points

$$\left(x_i^E, y_j^E\right), \left(x_{i+1}^E, y_j^E\right), \left(x_i^E, y_{i+1}^E\right), \text{ and } \left(x_{i+1}^E, y_{i+1}^E\right).$$

Specifically, the interpolation can be written in its explicit bilinear form

$$u = \frac{x' - x_i^E}{x_{i+1}^E - x_i^E}, \quad v = \frac{y' - y_j^E}{y_{j+1}^E - y_j^E} \tag{11}$$

$$f\left(x', y'\right) = \begin{bmatrix} 1-u & u. \end{bmatrix} \begin{bmatrix} f_{ij}^E & f_{i,j+1}^E \\ f_{i+1,j}^E & f_{i+1,j+1}^E. \end{bmatrix} \begin{bmatrix} 1-v \\ v. \end{bmatrix}. \tag{12}$$

This formulation ensures that the interpolated ETAD correction parameters t and τ preserve the original spatial variability while avoiding excessive smoothing artifacts.

Both the original ETAD raster and the target raster were generated through grid discretization, and the interpolation function was evaluated at each ETAD grid point to obtain the resampled ETAD layers. Subsequently, co-registration parameters derived from the interferometric processing of the reference image were applied to the selected ETAD correction layers to ensure pixel-level alignment

$$\begin{cases} r_E = r_o + \Delta r\left(r_{reference}, r_o\right) \\ a_E = a_o + \Delta a\left(a_{reference}, a_o\right) \end{cases} \tag{13}$$

where $r_{reference}$ and $a_{reference}$ denote the range and azimuth pixel coordinates of the reference image, r_o and a_o are the range and azimuth pixel coordinates of the ETAD image, and Δr and Δa represent the range and azimuth co-registration offsets, respectively.

After establishing spatial consistency between the ETAD and reference grids, we apply corrections to the range-direction timing error layers: TRO, ION, SET, and DOP. These components directly modulate the two-way signal travel time along the radar line of sight, thereby introducing slant-range path delays that map to interferometric phase and bias deformation velocity and height estimates. Their correction is therefore applied at the interferogram level—prior to PS detection, to suppress phase noise while preserving PS coherence. Azimuth-direction timing errors, they cause lateral displacement of pixel positions in the azimuth direction.

Next, the total time delays in range and azimuth directions are computed by summing the relevant ETAD layers. Special attention is paid to the sign conventions: the BIS azimuth correction and DOP-induced range shift correction are provided with opposite signs in the ETAD product. The corresponding spatial offsets are derived through the following equations:

$$\Delta l_{rg} = \frac{\Delta\tau_{rg} \cdot c}{2} \tag{14}$$

$$\Delta\tau_{rg} = \Delta\tau_{\text{tro}} + \Delta\tau_{\text{ion}} + \Delta\tau_{\text{geo,rg}} - \Delta\tau_{\text{dopper,rg}} \tag{15}$$

$$\Delta l_{az} = \Delta_\tau \cdot \upsilon_{beam} \tag{16}$$

$$\Delta_\tau = \Delta\tau_{\text{geo,az}} - \Delta\tau_{\text{bistatic}} + \Delta\tau_{\text{fm}} \tag{17}$$

where υ_{beam} is the zero-DOP beam velocity. The beam velocity of S-1 is in the order of 6800 m/s and is annotated in the ETAD product.

Then, for each interferogram, the timing information of the reference $\Delta\tau_{reference}$ and secondary images $\Delta\tau_{secondary}$, was derived from the interferometric processing parameters

$$\Delta\tau_{\text{reference}} = \left(\Delta\tau_{\text{TRO}} - \Delta\tau_{\text{ION}} + \Delta\tau_{\text{SET,rg}} + \Delta\tau_{\text{DOP}}\right)^{\text{reference}} \tag{18}$$

$$\Delta\tau_{\text{secondary}} = \left(\Delta\tau_{\text{TRO}} - \Delta\tau_{\text{ION}} + \Delta\tau_{\text{SET,rg}} + \Delta\tau_{\text{DOP}}\right)^{\text{secondary}}. \tag{19}$$

Based on the interferometric processing parameters, the differential signal path delay between the reference and secondary images was calculated for each interferogram

$$\Delta\tau_{\text{inf}} = \begin{cases} \Delta\tau_{\text{reference}} - \Delta\tau_{\text{secondary}}, & \text{if day}_{\text{reference}} > \text{day}_{\text{secondary}} \\ \Delta\tau_{\text{secondary}} - \Delta\tau_{\text{reference}}, & \text{otherwise.} \end{cases} \tag{20}$$

The differential delay $\Delta\tau_{\text{inf}}$ is then converted into phase offset ϕ_{ETAD} using the radar carrier wavelength λ

$$\phi_{\text{ETAD}} = -\Delta\tau_{\text{inf}} \cdot \frac{2\pi c}{\lambda}. \tag{21}$$

In SAR SLC interferometrythe phase measurement in an interferogram can be decomposed into several components as follows:

$$\phi_{\text{ori}} = -2\pi a + \phi_{\text{topo}} + \phi_{\text{defo}} + \phi_{\text{orb}} + \phi_{\text{atmo}} + \phi_{\text{noise}^*} \tag{22}$$

where a indicates the integer number of phase ambiguity, ϕ_{topo} is the topographic phase, ϕ_{defo} is the deformation phase, ϕ_{orb} is the orbital phase, ϕ_{atmo} is the atmospheric phase, and ϕ_{noise^*} is the noise phase.

The corrected phase is obtained by subtracting the ETAD-derived phase offset from the original interferometric phase

$$\phi_{\text{corrected}} = \phi_{\text{ori}} - \phi_{\text{ETAD}}. \tag{23}$$

The corrected interferogram is subsequently used for PS point identification and deformation inversion. Stable scatterers are identified based on the amplitude dispersion index

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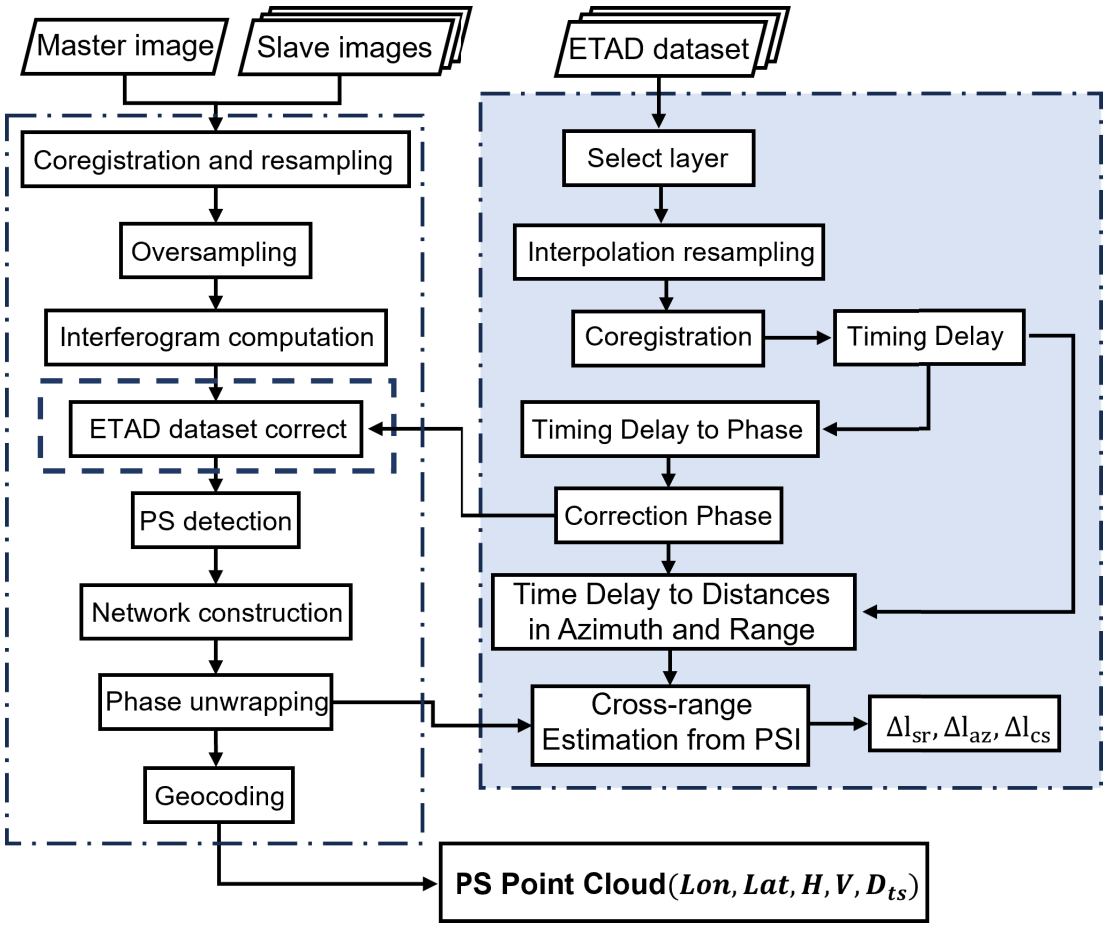


Fig. 2. Generalized workflow for incorporating ETAD corrections into PSI processing.

$D_A(< 0.25)$. The final parameter vector for each PS point is expressed as $[h_j, v, a]$, corresponding to height, linear velocity, and phase ambiguity, respectively.

In terms of PS points positioning correction. The ETAD product provides a set of correction layers, each of which introduces subtle timing adjustments either in the azimuth or slant-range dimension. Although these timing offsets are small—typically on the order of microseconds—they play a critical role in determining the final spatial positioning of radar targets on the ground. This section establishes the physical and mathematical linkage between these time-domain corrections and the corresponding geolocation shifts, forming the conceptual bridge from timing correction to spatial accuracy.

As illustrated in Fig. 3(a), azimuth information is derived through the synthetic aperture process. The key timing variable is the moment at which the reflected radar signal reaches the characteristic phase center—denoted as azimuth time t_{az}. If this time is perturbed by a correction $\Delta t_{az} = t'_{az} - t_{az}$, the resulting along-track displacement on the ground is

$$\Delta l_{az} = v_g \left(t'_{az} - t_{az}\right) \tag{24}$$

where v_g represents the projection speed of the satellite on the ground, and this offset only affects the horizontal positioning accuracy.

In the range direction [see Fig. 3(b)], the target's location is inferred from the two-way pulse travel time, or slant-range time $\tau_s r$. The geometric slant range is given by

$$l_{sr} = c\frac{\tau_{sr}}{2}. \tag{25}$$

A correction in range timing $\Delta\tau_{sr}$ leads to a slant-range Δl_{sr} and ground range Δl_{gr} shift

$$\Delta l_{sr} = c\frac{\tau'_{rg} - \tau_{rg}}{2} \tag{26}$$

and

$$\Delta l_{gr} = \frac{\Delta l_{sr}}{sin\theta}. \tag{27}$$

Here, θ is the local incident angle.

The cross-range distance ("elevation") of a point is estimated from at least one interferometric SAR observation as

$$\hat{l}_{cs} \approx c_0 + \frac{\lambda}{4\pi}\frac{l_{sr}}{B_\perp}\hat{\phi}_{corrected} \tag{28}$$

where c_0 is the cross-range (elevation) of a reference point, which is assumed to be known. The λ is the wavelength, $B_\perp$ is the perpendicular baseline of the interferometric pair, and $\hat{\phi}_{corrected}$ is the estimated unwrapped topographic phase relative to the reference point. Note that the assumption of a "known" reference point position is rarely satisfied in practice.

The cross-range l_{cs} together with azimuth l_{az}, and range l_{sr} complements the 3-D orthogonal radar coordinate system.

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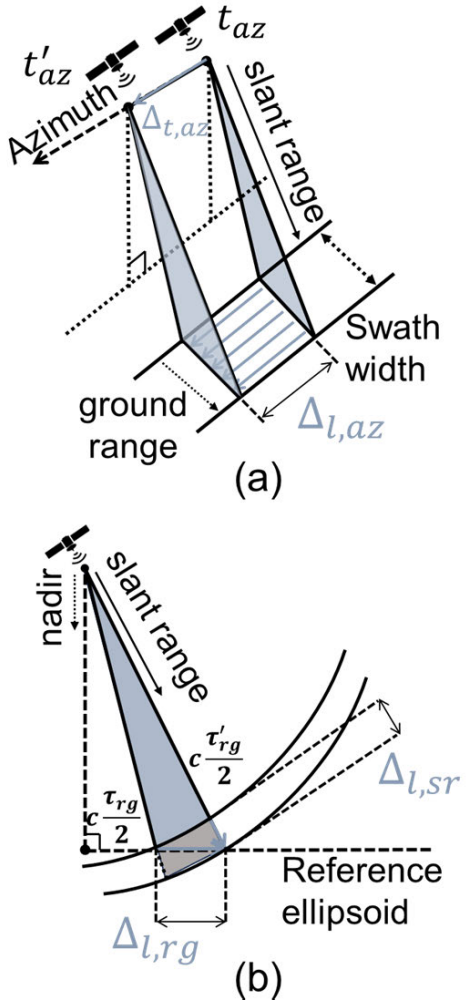


Fig. 3. Azimuth and range delays and their impact on geocoded coordinates. (a) Azimuth timing perturbation and the resulting along-track shift and (b) range timing perturbation and the resulting slant-range and ground-range shifts.

The corresponding position of point scatterer T, with state vector $\mathbf{T} = [x_T, y_T, z_T]$ in a 3-D TRF, is estimated using the range-DOP-ellipsoid equations.

For range, the geometric distance l_{sr} from scatterer T to satellite $\mathbf{S}(t_{az})$ is a function of the satellite state vectors and scatterer state vector

$$\|\mathbf{S}(t_{az}) - \mathbf{T}\| - l_{sr}^2 = 0 \tag{29}$$

where $\mathbf{S}(t_{az})$ is the satellite state vectors at the zero-DOP time of imaging target t_{az}.

For DOP, the scatterer T is viewed perpendicular to the orbit, i.e.,

$$f_D(t_{az}) + \frac{2}{\lambda}\frac{(\mathbf{S}(t_{az}) - \mathbf{T})}{|\mathbf{S}(t_{az}) - \mathbf{T}|}\mathbf{V}(t_{az}) = 0 \tag{30}$$

where $\mathbf{V}(t_{az})$ is the velocity vector of the satellite at the instant of imaging target T, and $f_D(t_{az})$ is the DOP frequency of scatterer T at azimuth position l_{az}.

Finally, for the ellipsoid, T has a vertical elevation $H(l_{cs})$ above a reference ellipsoid with semi-major and semi-minor axis m and f, respectively, hence,

$$\frac{x_T^2}{(m + H(l_{cs}))^2} + \frac{y_T^2}{(m + H(l_{cs}))^2} + \frac{z_T^2}{(f + H(l_{cs}))^2} - 1 = 0. \tag{31}$$

$H(l_{cs})$ is calculated from the cross-range position

$$H(l_{cs}) = l_{cs} \cdot \sin\theta_{\text{inc},T} \tag{32}$$

where $\theta_{\text{inc},T}$ is the incidence angle at T. Optionally, the 3-D TRF coordinates can be further transformed into a national/local coordinate system.

The precise 3-D geolocation relies on the precision of azimuth coordinate l_{az}, range coordinate l_{sr}, the estimated cross-range $\hat{l}_{cs}$, and the satellite orbit vectors.

The relative cross-range differences between the PS point cloud and the reference point are determined using PS interferometry. The estimates of these relative differences will improve when more data will be used in the interferometric stack.

A stack of $n+1$ acquisitions generates n independent pairs with different baselines $[B_{\perp,1}, \ldots, B_{\perp,n}]^T$ and phase observations $[\phi_{corrected,1}, \ldots, \phi_{corrected,n}]^T$ for target T, relative to the reference point. Using (28), the functional model with the initial range value of l_{sr}^0 can be written as

$$y = E\left\{\begin{bmatrix}\phi_{corrected,1}\\ \vdots \\ \phi_{corrected,n}\end{bmatrix}\right\} = \hat{G}_{n\times 1}\cdot l_{cs} = \begin{bmatrix}-\frac{4\pi \mathrm{B}_{\perp,1}}{\lambda l_{sr}^0}\\ \vdots \\ -\frac{4\pi \mathrm{B}_{\perp,n}}{\lambda l_{sr}^0}\end{bmatrix}\cdot l_{cs}. \tag{33}$$

The estimated cross-range and its precision $\sigma^2_{\hat{c}_T}$ are

$$\hat{l}_{cs} = \left(G^T Q_y^{-1} G\right)^{-1} G^T Q_y^{-1} y, \text{ and } \sigma^2_{\hat{l}_{cs}} = \left(G^T Q_y^{-1} G\right)^{-1}. \tag{34}$$

The matrix G is a diagonal matrix with entries $[-(4\pi \mathrm{B}_{\perp,1})/(\lambda \cdot \mathrm{l}_{\mathrm{sr}}^0), \ldots, -(4\pi \mathrm{B}_{\perp,\mathrm{n}})/(\lambda \cdot \mathrm{l}_{\mathrm{sr}}^0)]^{\mathrm{T}}$, l_{sr}^0 is the initial range value, and λ is the wavelength. The covariance matrix Q_y of time-series phase observations is a diagonal matrix with entries $[\sigma^2_{\phi_{corrected,1}}, \ldots, \sigma^2_{\phi_{corrected,n}}]^T$, σ_ϕ is the standard deviation (STD) of a single phase observation.

B. Assessment of ETAD Integration for PSI: Phase, Deformation, Time-Series, and Positioning

We develop a metric-driven framework to quantify how Sentinel-1 ETAD improves PSI, focusing on phase quality, deformation estimation accuracy, and PS geolocation performance.

For evaluating interferometric phase quality, two complementary indicators are employed: sum of phase differences (SPDs) and phase STD (PSD). SPD is a first-order statistical measure that quantifies the local smoothness of the interferometric phase. It is computed as the sum of absolute phase differences between each pixel and its neighboring pixels within a defined window. A smaller SPD value indicates that

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the phase varies smoothly in space, reflecting higher local phase consistency and better interferometric quality.

SPD is defined as the sum of the average phase difference (APD) at each pixel

$$SPD = \sum_{x=1}^{p} \sum_{y=1}^{q} APD(x, y) \tag{35}$$

where the APD at pixel (x, y) is given by

$$APD(x, y) = \frac{1}{8} \sum_{l=-1}^{1} \sum_{k=-1}^{1} |\phi(x, y) - \phi(x + l, y + k)|. \tag{36}$$

Here, $\phi(x, y)$ denotes the interferometric phase value at pixel (x, y); p and q are the number of rows and columns of the interferogram, respectively; and (l, k) are the relative coordinates of the eight neighboring pixels within a 3×3 window. The normalization factor $1/8$ ensures that the APD represents the mean difference between the central pixel and its neighbors.

PSD is a second-order statistical measure that characterizes the variability of the phase after removing a local linear phase trend. It is calculated as the STD of residual phases within a window, where the local linear phase ramp is estimated and subtracted. A smaller PSD value implies that the residual phase noise is lower, indicating higher local phase stability and coherence.

The PSD is calculated as

$$\text{PSD} = \sqrt{\frac{\sum_W \left(\phi_{W,ij} - \bar{\phi}_{W,ij}\right)^2}{W - 1}} \tag{37}$$

where $\bar{\phi}_{W,ij}$ is the local linear phase ramp over a rectangular window centered at the point. It is estimated by averaging the first differences of the interferometric phases in range and azimuth, respectively. W is the number of pixels in a window used for calculation. SPD emphasizes local phase continuity (sensitive to abrupt spatial phase changes), PSD emphasizes phase stability (sensitive to random noise and decorrelation).

For deformation estimation, a comparative analysis is conducted across four processing strategies: 1) uncorrected PSI; 2) PSI with spatiotemporal filtering; 3) PSI with GACOS-based TRO correction; and 4) the proposed ETAD-based correction method.

For evaluating the accuracy of deformation rate estimation, the STD of the estimated rates is utilized as a quantitative metric, where a lower STD indicates higher consistency and reduced uncertainty in the deformation results.

For evaluating time-series deformation estimation, GNSS observations are employed as independent References. Two statistical indicators, mean absolute error (MAE) and root mean square error (RMSE), are calculated to quantify the accuracy. The MAE and the mse are defined as

$$MAE = \frac{1}{N} \sum_{i=1}^{N} \left| d_i^{\text{InSAR}} - d_i^{\text{GNSS}} \right| \tag{38}$$

$$RMSE = \sqrt{\frac{1}{N} \sum\nolimits_{i=1}^{N} \left(d_i^{\text{InSAR}} - d_i^{\text{GNSS}}\right)^2}. \tag{39}$$

Here, d_i^{InSAR} denotes the deformation measurement obtained from the InSAR time series at epoch i, and d_i^{GNSS} is the corresponding reference deformation derived from GPS observations. N represents the total number of temporal samples used for the comparison.

The MAE reflects the average magnitude of deviations between the two datasets, providing a measure of overall accuracy. In contrast, the RMSE gives greater weight to large deviations due to its quadratic form, and thus better captures the consistency and reliability of the InSAR-derived displacement time series with respect to the GNSS benchmark. In both cases, smaller MAE and RMSE values indicate higher agreement with the GNSS observations, thus reflecting better accuracy and reliability of the time-series deformation estimates.

We employ two complementary indicators to evaluate the quality of deformation time series. The first is the STD of displacements (SDDs), defined as the STD of residuals between the estimated deformation model and the observed time series

$$SDD = \sqrt{\frac{1}{S - 1} \sum\nolimits_{s=1}^{S} (e_s - \bar{e})^2}. \tag{40}$$

Here, S is the total number of epochs, e_s is the residual displacement at epoch s obtained as the difference between the observed displacement and the model-predicted displacement, and $\bar{e}$ denotes the mean of the residuals. A smaller SDD indicates a closer agreement between the time series and the deformation model, reflecting lower noise and higher reliability of the InSAR-derived displacements, while a larger SDD suggests the presence of atmospheric artifacts, unwrapping errors, or unmodeled deformation components.

The second indicator is the spatiotemporal consistency (STC), which evaluates the noise level in the deformation time series and is independent of the deformation model. It is defined as the minimum RMSE of the double-differences between a PS i and its surrounding PS j within a specified search radius

$$STC = \min \left\{ \frac{\lambda}{4\pi} \sqrt{\frac{1}{S - 1} \sum\nolimits_{s=1}^{S-1} \left(\left(\varphi_j^{0s} - \varphi_i^{0s}\right) - \left(\varphi_j^{0(s+1)} - \varphi_i^{0(s+1)}\right) \right)^2 } \right\}. \tag{41}$$

Here, φ_i^{0s} and φ_j^{0s} denote the interferometric phases at PS i and its neighboring PS j at epoch s, respectively. The surrounding PS j are selected within a given spatial search radius (typically 50–250 m), and the minimum RMSE across all available neighbors is taken to suppress the influence of deformation signals, leaving a measure that primarily reflects the measurement noise. Together with the SDD, which measures model conformity, the STC provides a complementary perspective by capturing noise characteristics, and the two metrics jointly offer a comprehensive assessment of PSI time-series quality.

To assess the impact of ETAD on geometric positioning, the geocoding accuracy of the corrected PS points is then independently validated using high-resolution airborne LiDAR point cloud data. This comparison allows us to quantify the spatial alignment improvement and evaluate the contribution of ETAD-derived timing corrections to accurate PS localization.

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TABLE I
SAR DATA CHARACTERISTICS

Satellite/Parameter	Sentinel-1
Frame	509
Path	62
Band[cm]	X[3.1]
Start Date	2023.08.11
End Date	2024.09.10
Number of images	33
Acquisition mode	IW
Pass direction	Desc
Polarization	VV/VH
PRF[Hz]	1717.13
RSR[MHz]	64.35
Slant range spacing[m]	2.33
Azimuth spacing[m]	13.97
Rang. Bandw. [MHz]	56.5
Azim. Bandw. [Hz]	327

This comprehensive evaluation framework provides a robust basis for assessing how ETAD can enhance both the geometric fidelity and deformation reliability of PSI-derived results, paving the way for improved geodetic applications using Sentinel-1.

IV. EXPERIMENT SETUP

This study investigates the integration of Sentinel-1 ETAD corrections into a standard PSI processing chain, with the objective of improving interferogram quality, deformation time-series accuracy, and PS geolocation precision. To demonstrate and evaluate the effectiveness of the proposed approach, we selected the central urban area of Kunming, China, as the test site. The area is characterized by heterogeneous urban infrastructure and moderate terrain variation, which together pose challenges for phase stability and geolocation accuracy. A total of 33 Sentinel-1 IW mode SLC images were acquired in descending orbit, spanning the period from August 2023 to September 2024. The spatial extent of the study area is shown in Fig. 1, and the key acquisition parameters are summarized in Table I.

To rigorously evaluate the impact of correction strategy on PSI deformation estimates, we conduct a four-way comparative experiment within an identical StaMPS processing chain (same reference image, PS selection, unwrapping, and inversion parameters).

1) PSI+ETAD applies ETAD's sensor-level timing corrections—to generate timing-corrected interferograms before PSI estimation, thereby addressing TRO, ION, SET, and DOP-related delays in a geometrically consistent manner.

2) PSI+GACOS employs GACOS as an atmospheric phase screen estimator: its zenith total delay (ZTD) fields—generated by fusing GNSS observations with ECMWF ERA5 data and explicitly decomposed into stratified (topography-correlated) and turbulent (localized, rapidly varying) components—are projected to the line of sight and applied to individual interferograms prior to unwrapping.

3) PSI+STF implements spatiotemporal filtering—representing the empirical, data-driven noise suppression widely adopted in operational PSI when no external calibration is available.

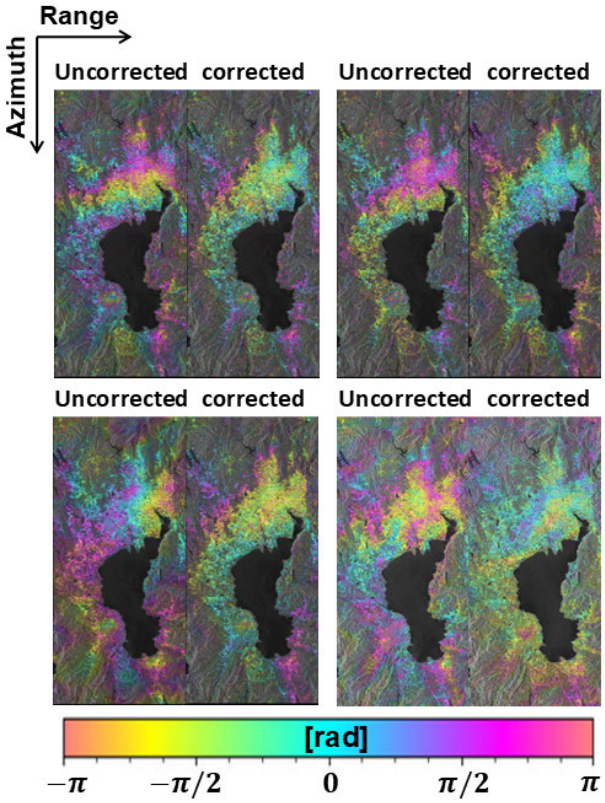


Fig. 4. Interferogram comparison: with and without ETAD correction.

4) PSI (uncorrected) serves as the baseline, with no explicit correction for atmospheric, timing, or geometric errors. By controlling all processing variables except the correction method, differences in deformation velocity STD, time-series stability (SDD and STC), and geolocation residuals reflect solely the intrinsic effectiveness of each strategy.

In addition, high-resolution airborne LiDAR data were collected over two subregions—namely, the Kunming Convention and Exhibition Center and Caiyun North Road—to serve as independent ground truth for geolocation validation. The LiDAR dataset was acquired using the Leica CityMapper-2S system equipped with a Hyperion2 laser operating at 200 kHz. The resulting point cloud achieves an average density exceeding 2.05 points per square meter, with lateral overlap between 14.1% and 57.0%. The height accuracy meets the requirements of a 1:2000 topographic survey, with mean vertical errors better than 0.20 m. These high-precision data offer a reliable benchmark to assess the positioning accuracy of PS points after ETAD correction.

V. RESULTS

This section systematically presents and compares the PSI results obtained before and after the integration of ETAD corrections. The analysis covers four key aspects: interferometric phase quality, linear deformation velocity estimation, deformation time-series reconstruction, and the refinement of PS geolocation accuracy. Through a comprehensive, multidimensional comparison against baseline solutions, the effectiveness of the ETAD-integrated processing chain is validated,

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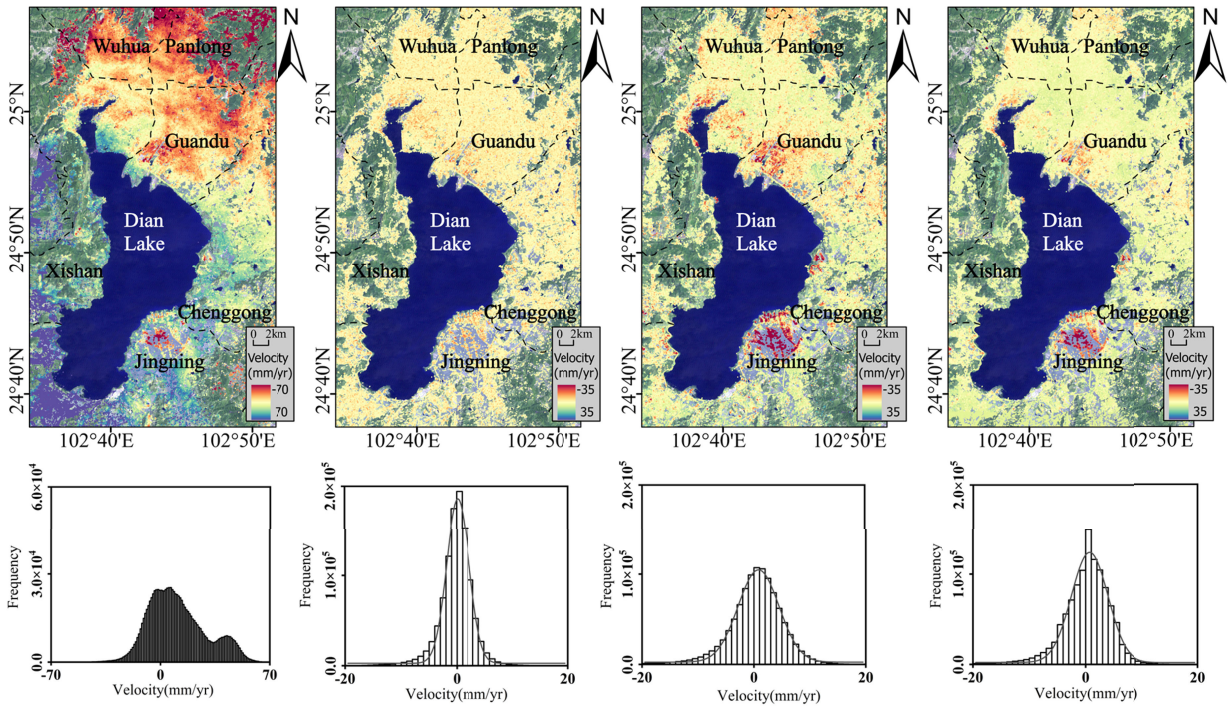


Fig. 5. Velocity maps and histograms for four methods: PSI (uncorrected), PSI+STF, PSI+GACOS, and PSI+ETAD.

TABLE II
INTERFEROGRAM QUALITY ASSESSMENT WITH SPD (LOCAL PHASE SMOOTHNESS) AND PSD (RESIDUAL PHASE VARIABILITY) BEFORE AND AFTER ETAD CORRECTION

Interferogram Type	SPD	PSD
Original interferograms	18.94	1.875
Corrected interferograms	7.43	1.392
Relative improvement (%)	60.77%	40.30%

demonstrating its capability to significantly improve the precision, stability, and reliability of PSI-based deformation monitoring and geolocation.

A. Interferogram Quality

This section evaluates the impact of ETAD corrections on interferometric phase quality. Both visual inspection and quantitative analysis are employed to demonstrate how ETAD-derived timing corrections enhance interferogram coherence and phase stability, which are critical prerequisites for accurate deformation retrieval in PSI processing.

Fig. 4 compares an original interferogram without correction against the corresponding result after applying ETAD-based phase delay corrections. Although fringe patterns vary across interferometric pairs, the corrected interferograms consistently show reduced complexity and exhibit smoother, more continuous phase patterns. These visual improvements highlight the enhanced structural clarity and interpretability of the interferometric signals.

To quantitatively assess these improvements, two widely used phase quality metrics are adopted: the SPD and the PSD, as summarized in Table II. SPD measures local phase continuity and is particularly sensitive to abrupt spatial variations, while PSD characterizes overall phase stability by capturing random noise and decorrelation effects. Together, these indicators provide a robust basis for evaluating phase quality before and after ETAD correction.

According to the results in Table II, ETAD-based phase correction substantially improves interferometric quality. The SPD decreases by 60.77% relative to the uncorrected case, demonstrating effective suppression of cumulative phase noise and a corresponding enhancement of interferometric coherence, which is essential for reliable PSI processing. The PSD is reduced by 40.3%, reflecting a significant smoothing of the interferometric signal and improved phase continuity across the scene.

B. PSI Product: Linear Displacement Velocity

Sections V-B–V-D evaluate the impact of ETAD corrections on PSI products, including linear displacement velocity, time-series displacements, and PS geolocation precision. This section focuses on linear deformation velocity and examines how ETAD-derived timing corrections influence the accuracy and reliability of PSI velocity fields. We compare four PSI results: 1) PSI (uncorrected); 2) PSI+STF (PSI with spatiotemporal filtering); 3) PSI+GACOS (PSI with GACOS-based TRO delay correction); and 4) PSI+ETAD (PSI with Sentinel-1 ETAD integration). It should be noted that, except for the STF method, no filtering was applied in the other approaches, such that the comparisons primarily reflect the effects of noise reduction versus interferogram-based corrections.

Fig. 5 presents the deformation velocity maps for the four variants. The PSI (uncorrected) result is strongly affected by long-wavelength phase artifacts, yielding a high STD

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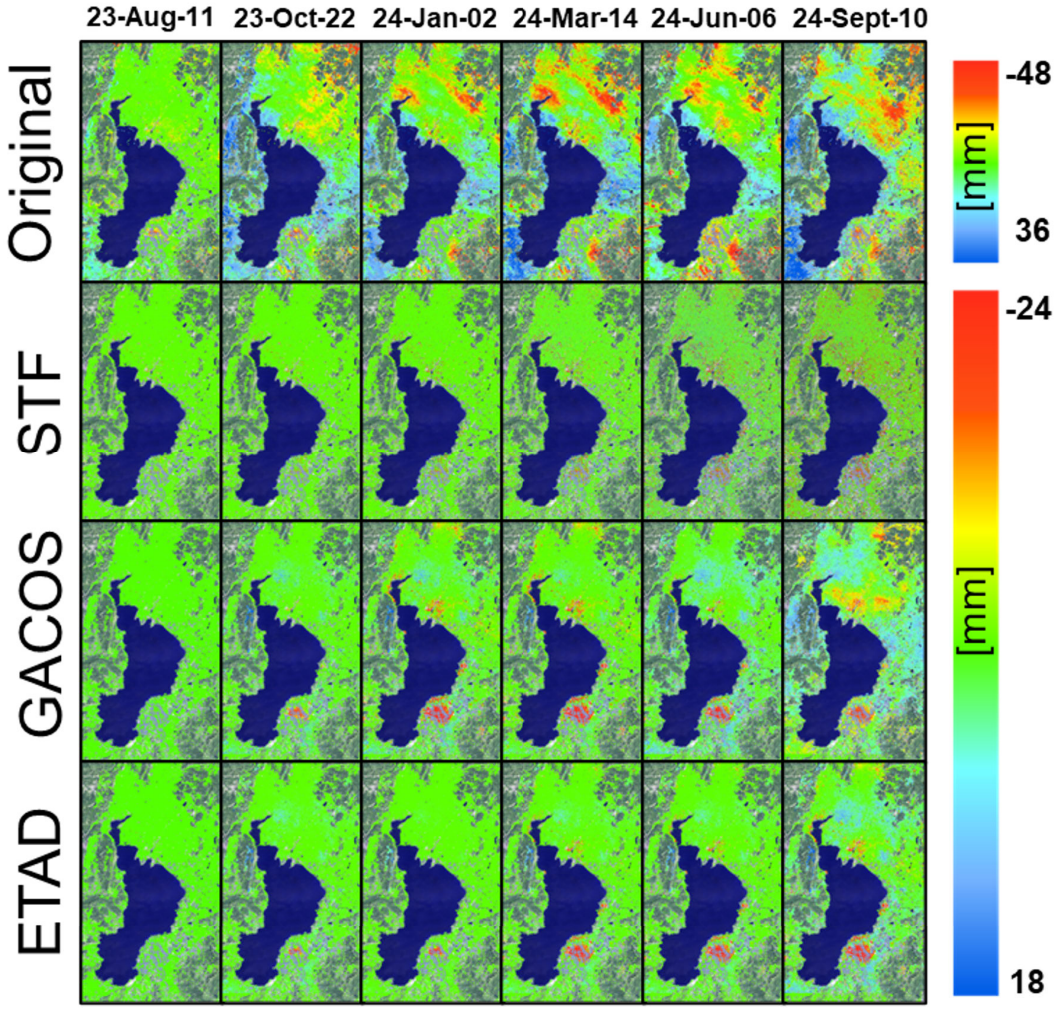


Fig. 6. Deformation time-series results from four methods: PSI (uncorrected), PSI+STF, PSI+GACOS, and PSI+ETAD.

of 18.64 mm/year and obscuring true deformation signals. PSI+STF suppresses 84.87% of the noise, reducing the STD to 2.82 mm/year; however, the improvement comes at the cost of over-smoothing, which may attenuate localized or subtle deformation features. PSI+GACOS lowers the STD to 6.34 mm/year (a 65.99% reduction), alleviating part of the atmospheric contribution but leaving residual noise in several areas. In contrast, PSI+ETAD achieves a more favorable balance between noise suppression and signal preservation: it reduces 71.78% of nondeformation noise relative to the uncorrected case—improving upon PSI+GACOS by an additional 5.79%—and yields a refined STD of 5.26 mm/year. These results demonstrate the effectiveness of ETAD integration in enhancing the precision and robustness of PSI-derived deformation velocity estimates.

From the distribution histograms, assuming that deformation rates within a localized region follow a Gaussian distribution, all three correction methods significantly improve the distributional properties of the velocity estimates. Among them, the spatiotemporal filtering and ETAD-based approaches exhibit superior correction performance, producing distributions that more closely approximate the expected normal behavior, thereby enhancing the interpretability of PSI-derived velocity products.

From the distribution histograms, and under the assumption that deformation rates within localized regions follow a Gaussian distribution, all three corrected PSI (PSI+STF, PSI+GACOS, and PSI+ETAD) substantially improve the statistical properties of the velocity estimates compared to the uncorrected PSI results. Among them, PSI+STF and PSI+ETAD achieve the best performance, yielding distributions that more closely approximate normality and thereby enhancing the interpretability and reliability of PSI-derived velocity products.

The subsidence patterns observed across the Kunming urban area are consistent with the region's well-documented geological setting. Extensive lacustrine soft clay deposits underlie the area, characterized by high compressibility and low bearing capacity [16]. Decades of rapid urban expansion, industrial activities, and municipal infrastructure development have induced significant fluctuations in groundwater levels [33]. Combined with historical groundwater extraction, these changes have altered pore-water pressures and redistributed effective stress within the subsurface, promoting gradual con-

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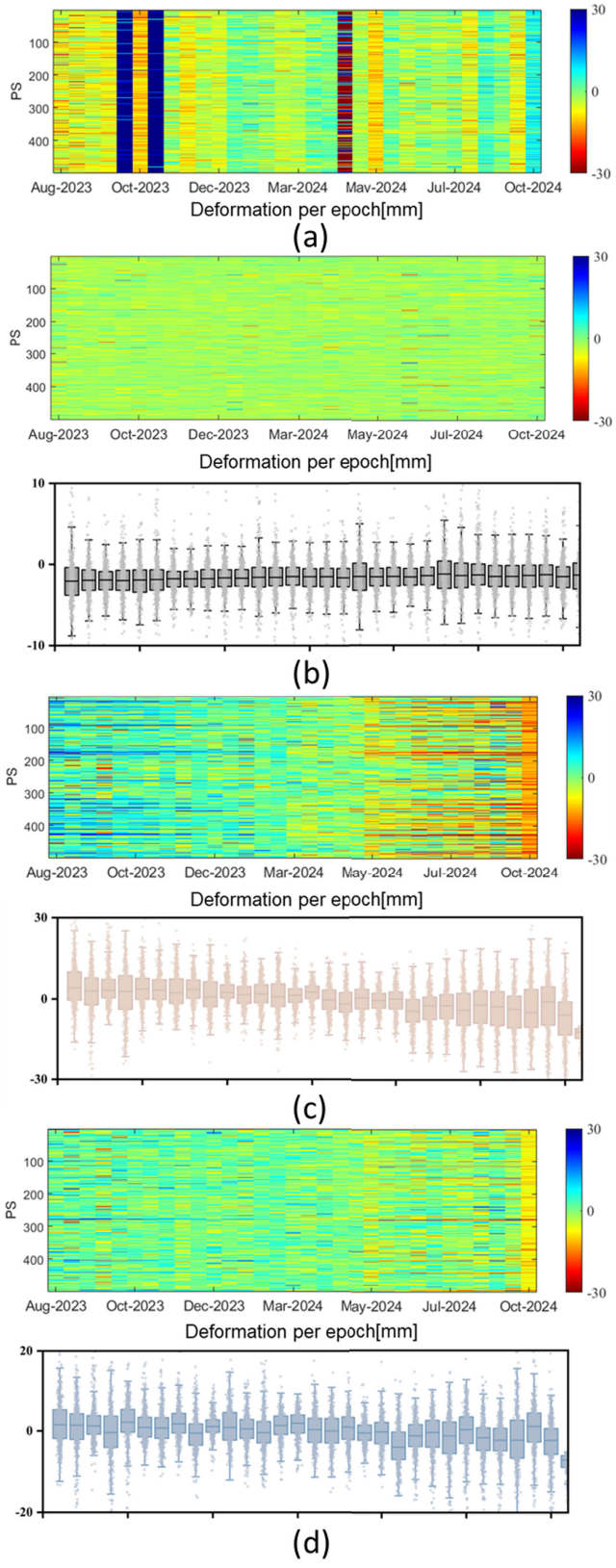


Fig. 7. Displacement time series of 500 randomly selected PS points in a high-deformation area, with residuals and box-plot statistics for different correction methods. (a) Uncorrected PSI, (b) PSI+STF, (c) PSI+GACOS, and (d) PSI+ETAD.

solidation of the soft sediments and resulting in sustained, low-rate subsidence [15]. This geomechanical process explains

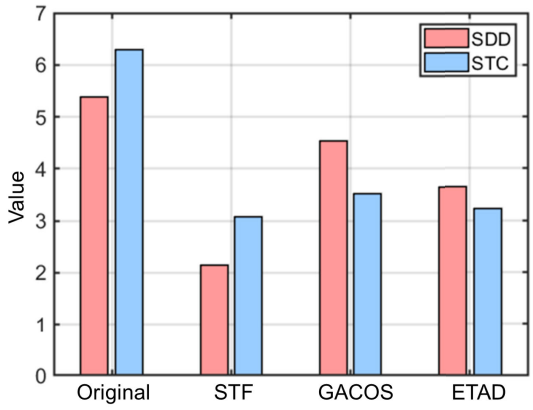


Fig. 8. Time-series quality assessment using SDD (model conformity) and STC (noise level): comparative analysis of PSI (uncorrected), PSI+STF, PSI+GACOS, and PSI+ETAD methods.

both the magnitude and spatial continuity of the deformation signal—making Kunming an ideal test site for evaluating PSI correction efficacy: strong, physically grounded signals allow subtle improvements in accuracy (e.g., reduced velocity std from 5.26 to 4.98 mm/yr in high-clay zones) to be robustly detected and attributed to ETAD integration.

Based on previous studies on surface deformation monitoring in Kunming, the results indicate that land subsidence exhibits pronounced spatial heterogeneity and strong anthropogenic driving factors. In general, subsidence in Kunming shows a clear spatial pattern, with more severe subsidence along the southern Dianchi Lake region and relatively stable conditions in the northern areas. Significant subsidence zones are mainly concentrated in rapidly urbanizing districts, including Xishan, Guandu, and Chenggong. Further localized investigations have reported evident subsidence around the Kunming International Convention and Exhibition Center, newly developed large-scale residential communities, and the Xiaobanqiao area, suggesting that intensive urban construction activities and ground disturbance play an important role in surface deformation. In addition, subsidence in the eastern and northeastern Dianchi Lake regions has been closely associated with underground resource exploitation and groundwater extraction, which have induced notable surface subsidence processes [28], [29], [34], [35].

Existing studies consistently suggest that land subsidence in Kunming is primarily driven by anthropogenic activities, such as urban development, groundwater withdrawal, and subsurface resource exploitation, while being strongly controlled by lacustrine soft soil deposits.

C. PSI Product: Time-Series Displacements

This section evaluates the effectiveness of ETAD corrections in improving the quality of PSI-derived deformation time series. The analysis investigates the extent to which atmospheric and systematic noise can be suppressed while retaining deformation signals, using comparative results from four processing variants: PSI (uncorrected), PSI+STF, PSI+GACOS, and PSI+ETAD.

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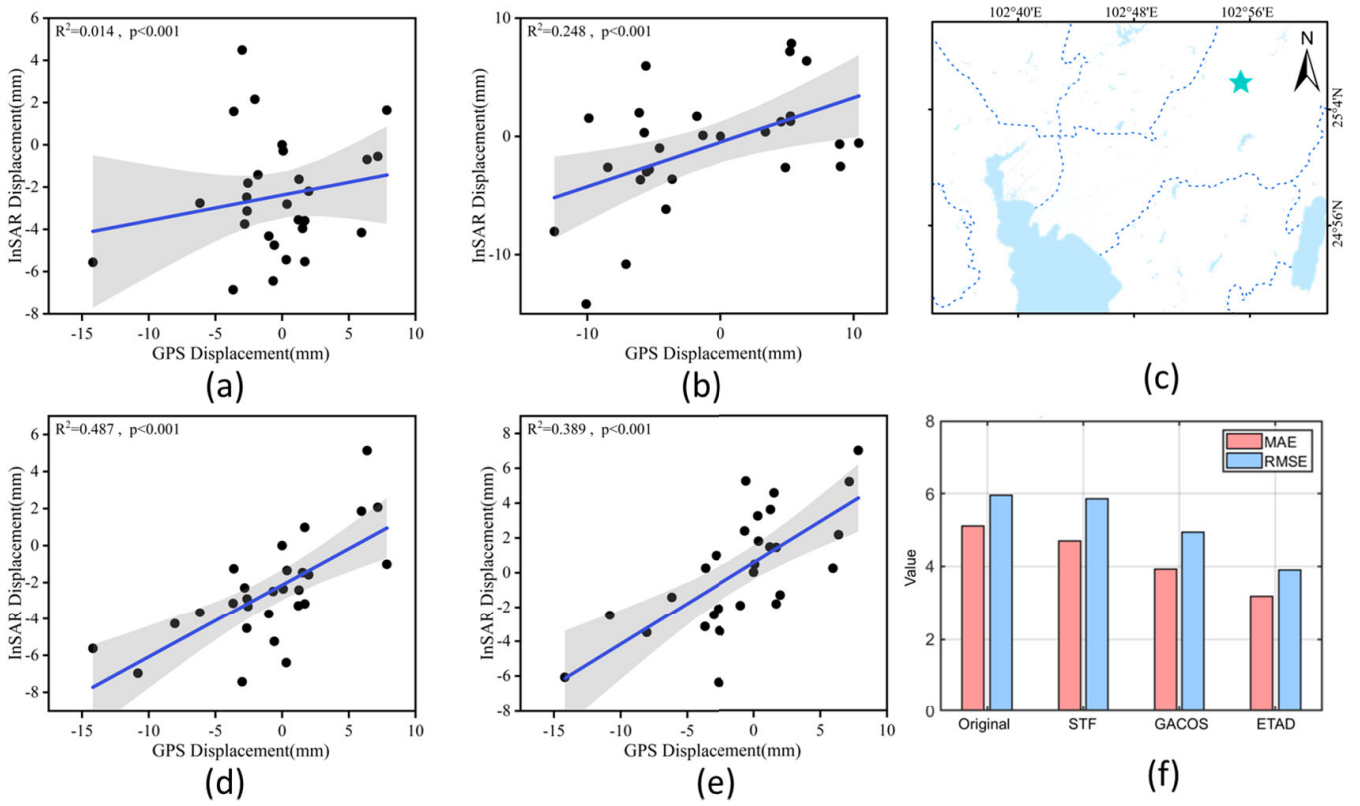


Fig. 9. GNSS-based validation of time-series deformation: (a)–(e) show comparisons between GNSS observations and InSAR results from (a) PSI (uncorrected), (b) PSI+STF, (d) PSI+GACOS, and (e) PSI+ETAD. (c) Location of the GNSS station, a continuously operating CORS site. (f) Accuracy metrics (MAE and RMSE) across the four methods.

As shown in Fig. 6, the uncorrected PSI time series exhibits strong temporal fluctuations, with severe noise obscuring the underlying deformation trend. The PSI+STF reduces these fluctuations but also attenuates deformation signals, particularly in regions with complex dynamics. The PSI+GACOS provides moderate improvement by compensating for part of the atmospheric delay, yet residual noise remains evident. In contrast, the PSI+ETAD results demonstrate superior temporal consistency, effectively suppressing high-frequency noise while preserving the displacement trend. These observations underscore the potential of ETAD integration to enhance both the stability and interpretability of PSI-derived time series.

To further assess the statistical characteristics of noise suppression across different correction strategies, 500 PS points were randomly selected from deformation-active regions. Fig. 7 presents the displacement time series for each correction method, together with their differences relative to the uncorrected PSI results, complemented by box plots that illustrate the residual distributions.

Fig. 7(a) shows the displacement time series from the uncorrected PSI results, where substantial noise is evident at several epochs, manifesting as abrupt temporal fluctuations. Fig. 7(b) presents the PSI+STF variant after conventional spatiotemporal filtering. While this approach effectively suppresses random noise, it also causes noticeable amplitude compression; the removed component resembles a near-constant offset, suggesting potential loss of meaningful deformation signals. Fig. 7(c) illustrates the PSI+GACOS results, corrected using GACOS TRO data, together with residuals relative to the uncorrected sequence. This method attenuates large-scale atmospheric artifacts while retaining the overall deformation magnitude. However, sudden phase jumps remain at certain epochs, and the residuals display temporal trends, indicating limited capability in handling localized or dynamic atmospheric variations. Fig. 7(d) presents the displacement time series after PSI+ETAD correction. Compared with the other strategies, PSI+ETAD achieves a more balanced performance, preserving deformation patterns while effectively reducing both random and systematic errors. The residuals removed by ETAD share a similar distribution pattern with those from PSI+GACOS, but with improved temporal consistency and fewer artifacts.

To quantitatively evaluate the quality of deformation time series, two complementary indicators are employed: the SDD and the STC. Fig. 8 summarizes the results for the four PSI variants. The uncorrected PSI data exhibit the poorest performance, with an average SDD of 5.34 and an STC of 4.12, indicating weak model conformity and high noise levels. The PSI+STF approach substantially improves both metrics, reducing SDD to 2.14 and STC to 3.07, though at the cost of signal distortion. The PSI+GACOS method yields an SDD of 4.53 and an STC of 2.76, reflecting moderate mitigation of atmospheric effects. In contrast, PSI+ETAD

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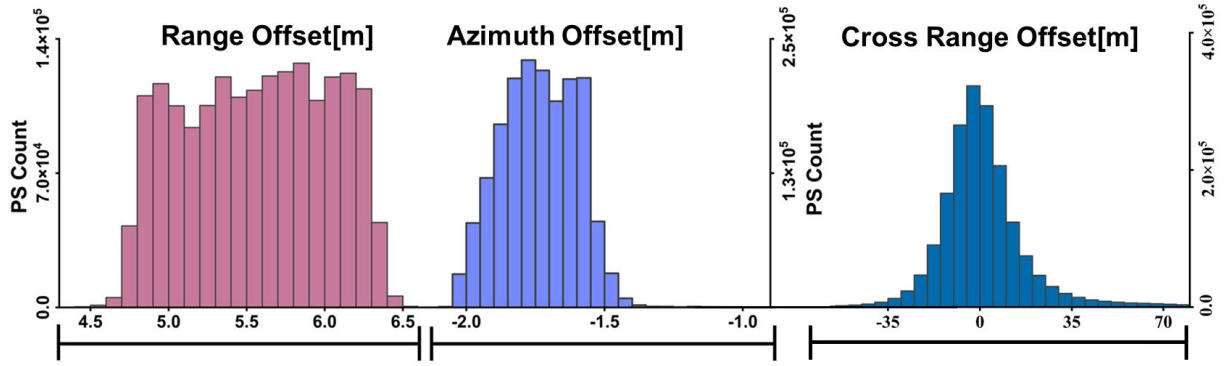


Fig. 10. Histograms of range, azimuth, and cross-range offsets induced by ETAD corrections.

achieves an SDD of 3.65 and an STC of 2.23, offering a more balanced tradeoff between improved model conformity and effective noise suppression.

To validate the deformation time series, high-precision GNSS observations are used as independent benchmarks. Fig. 9 compares the four PSI variants in terms of MAE and RMSE. The uncorrected PSI exhibits the poorest performance, with MAE of 4.69 mm and RMSE of 5.86 mm, reflecting substantial deviations from GNSS data, pronounced temporal instability, and mismatched trends. The PSI+STF variant reduces temporal fluctuations but also attenuates deformation signals, leading to errors of 5.12 mm (MAE) and 5.96 mm (RMSE). The PSI+GACOS method achieves moderate improvement, with MAE reduced to 3.92 mm and RMSE to 4.95 mm, aligning more closely with GNSS trends yet still retaining short-term fluctuations.

In contrast, the PSI+ETAD results show the best agreement with GNSS, characterized by a smooth trajectory and improved consistency, with the lowest errors of 3.18 mm (MAE) and 3.91 mm (RMSE). These results confirm that ETAD corrections effectively suppress atmospheric and systematic noise while preserving deformation signals, thereby providing the most reliable PSI-derived displacement estimates for geodetic applications.

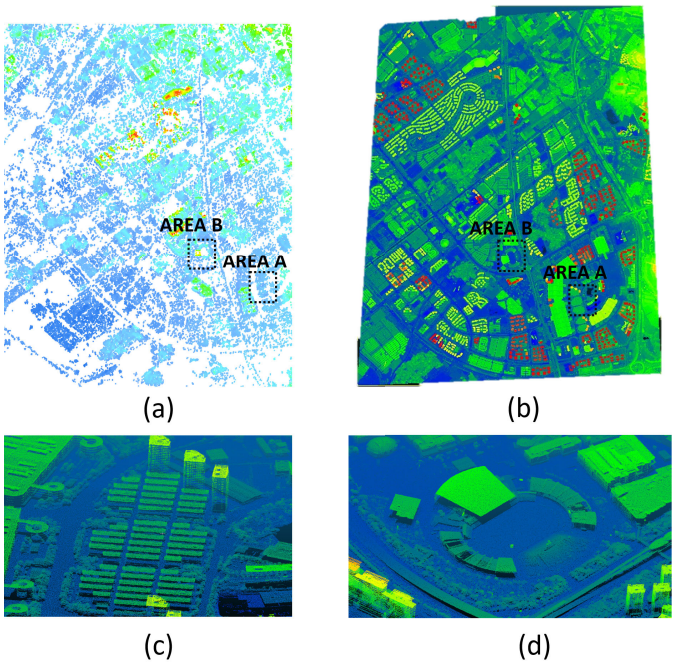


Fig. 11. Comparison of airborne LiDAR data and InSAR-derived PS point clouds over selected areas. (a) Distribution of PS point cloud, color-coded by elevation. (b) Corresponding airborne LiDAR point cloud. (c) Enlarged view of Area A. (d) Enlarged view of Area B.

D. PSI Product: Geolocation Precision

This section evaluates the impact of ETAD corrections on the geolocation accuracy of PS points. As a timing correction product, ETAD directly modifies the azimuth and slant-range positions of scatterers, thereby shifting their estimated 3-D locations. In addition, because ETAD also compensates for second-order phase errors, it can influence height estimation, which further affects the final geocoded positions of PS points in 3-D space.

To comprehensively assess geolocation accuracy, we first analyze the azimuth and range offsets within the radar coordinate system, as these represent the most direct effects of ETAD corrections. We then evaluate the resulting 3-D positioning shifts by comparing the corrected PS point cloud with high-resolution airborne LiDAR data. This analysis is conducted separately for horizontal and vertical components in order to quantify the effectiveness of ETAD in enhancing geolocation precision.

The distribution of azimuth and range offsets induced by ETAD corrections is shown in Fig. 10. Statistically, the mean slant-range offset is 5.566 m with a STD of 0.456 m, while the mean azimuth offset is 1.736 m with a STD of 0.135 m. The cross-range offset has a mean of 0.974 m and a STD of 16.509 m.

These results indicate that the range correction exceeds one slant-range pixel, whereas the azimuth correction remains below one azimuth pixel. The range correction is substantially larger than the azimuth correction, which is consistent with the physical nature of ETAD layers. Most delay components—particularly atmospheric effects such as TRO and ION delays—primarily influence the signal path length and thus introduce more significant offsets in the slant-range

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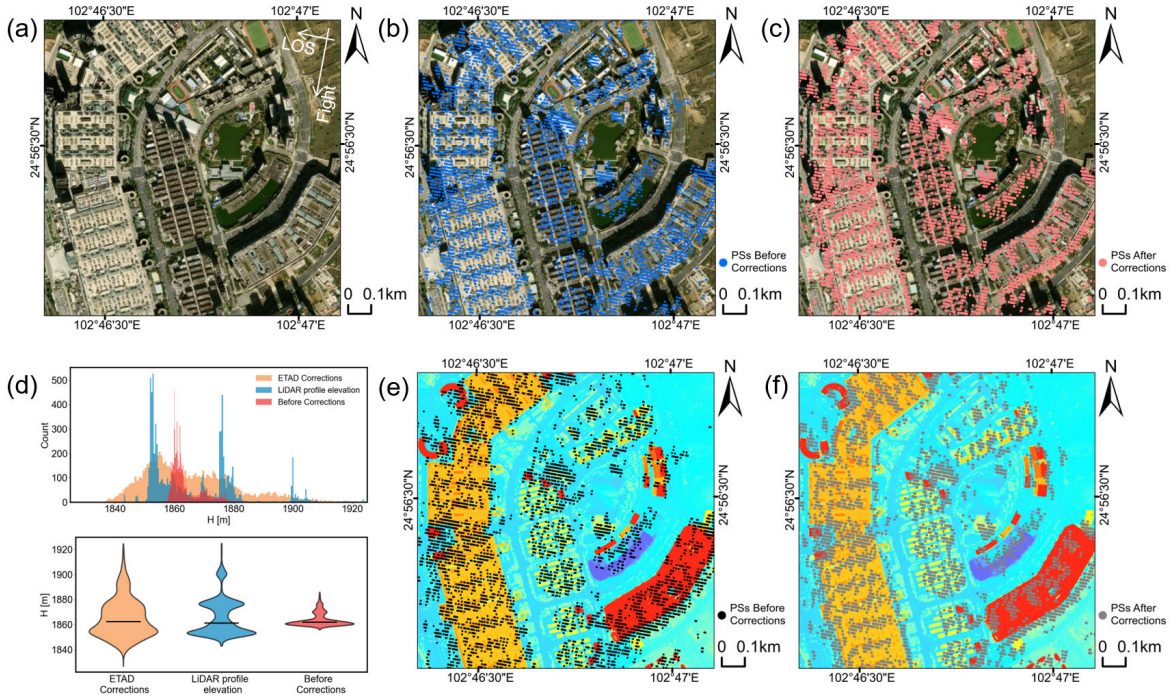


Fig. 12. Validation of PSI+ETAD geolocation improvement: (a) optical basemap (Google Earth) of Study Area A, with white arrow indicating satellite flight direction; (b) and (c) PS point planimetric distribution (range + azimuth) overlaid on the optical basemap before (b) and after (c) PSI+ETAD correction; (d) cross-range (elevation) histograms showing reduced spread and improved alignment with LiDAR-derived terrain after correction; and (e) and (f) PS points overlaid on airborne LiDAR point cloud before (e) and after (f) correction—demonstrating enhanced consistency.

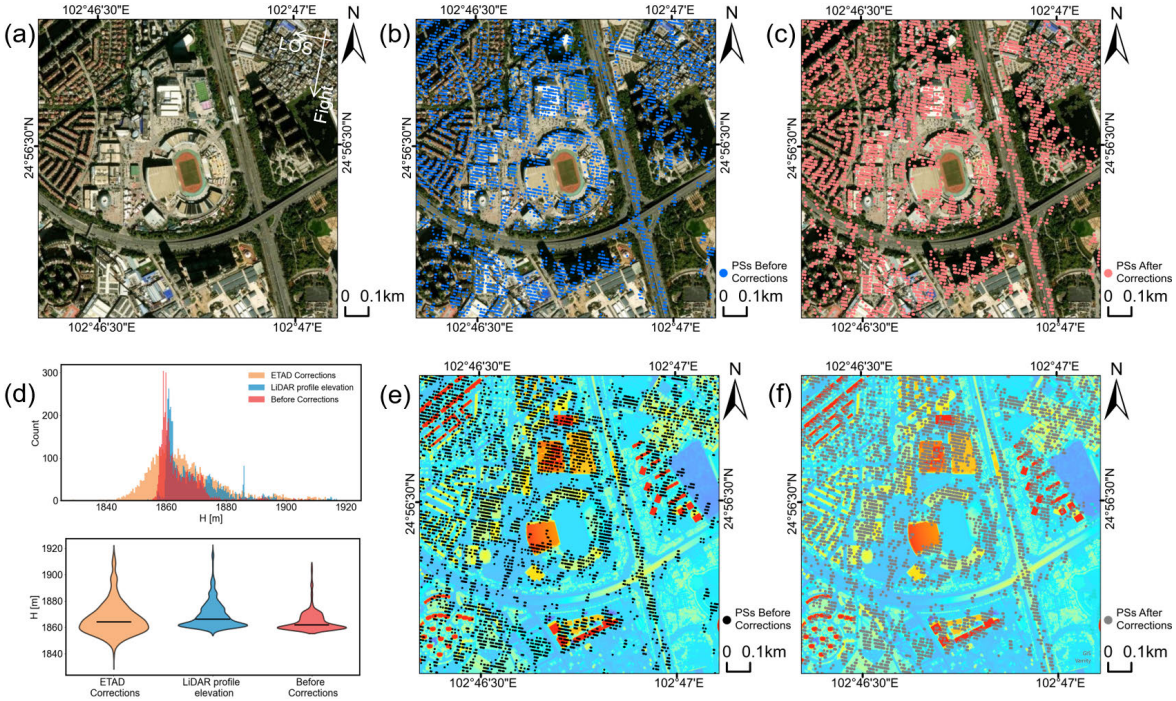


Fig. 13. Validation of PSI+ETAD geolocation improvement: (a) optical basemap (Google Earth) of Study Area B, with white arrow indicating satellite flight direction; (b) and (c) PS point planimetric distribution (range + azimuth) overlaid on the optical basemap before (b) and after (c) PSI+ETAD correction; (d) cross-range (elevation) histograms showing reduced spread and improved alignment with LiDAR-derived terrain after correction; and (e) and (f) PS points overlaid on airborne LiDAR point cloud before (e) and after (f) correction—demonstrating enhanced consistency.

dimension. By contrast, azimuthal shifts are generally smaller in magnitude but remain nonnegligible for precise geolocation tasks.

To further evaluate the effectiveness of ETAD-based geolocation correction, a comparative analysis was performed using high-resolution airborne LiDAR point cloud data as a bench-

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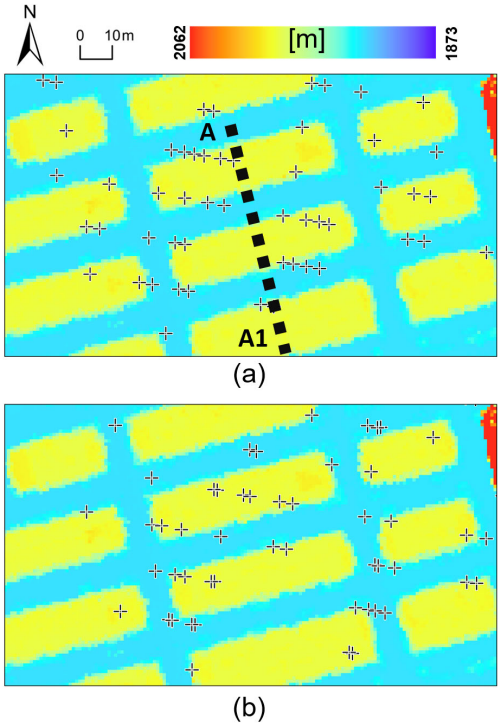


Fig. 14. 2-D horizontal geolocation comparison in Area A (building complex): PS point clouds (a) before and (b) after ETAD correction overlaid on LiDAR DSM.

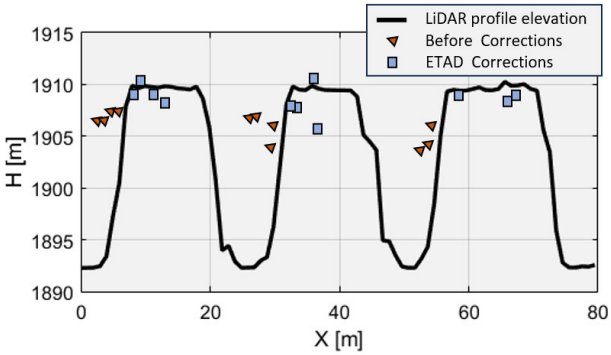


Fig. 15. 1-D vertical profile comparison in Area A along the AA′ PS point clouds before and after ETAD correction superimposed on LiDAR elevation.

mark. Two representative urban areas within the study site were selected for detailed examination, both covered by Sentinel-1 PSI results and high-density LiDAR measurements. Fig. 11 provides an overview of the distribution of the available LiDAR data and the corresponding InSAR-derived PS point clouds across the full study area. Figs. 12 and 13 show the spatial distributions of PS locations in the two study areas before and after correction. Compared with the uncorrected case, the corrected PS points exhibit more coherent spatial patterns and improved geometric consistency, indicating enhanced positioning performance at the area-wide scale.

In Area A, located near the building complex along Caiyun North Road, both horizontal and vertical positioning performance were examined. Fig. 14 presents the top-view overlay of the PS point cloud before and after ETAD correction on the LiDAR-derived digital surface model (DSM). Prior to correction, the PS points show significant lateral offsets

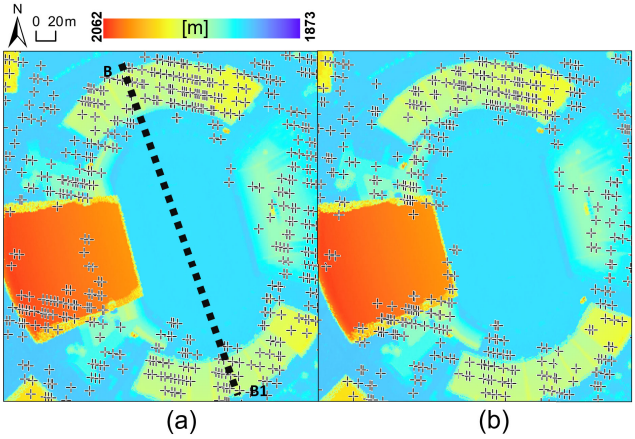


Fig. 16. 2-D horizontal geolocation comparison in Area B (stadium): PS point clouds (a) before and (b) after ETAD correction overlaid on LiDAR DSM.

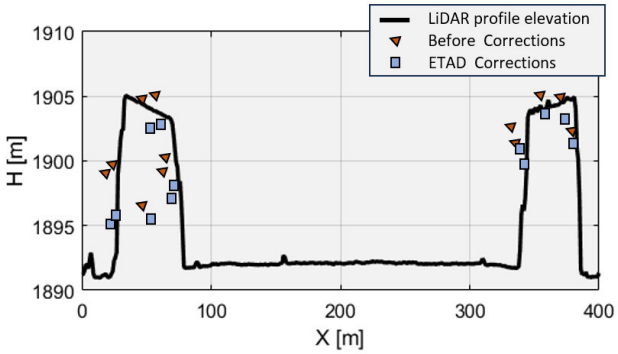


Fig. 17. 1-D vertical profile comparison in Area B along the BB′ PS point clouds before and after ETAD correction superimposed on LiDAR elevation.

relative to building edges. After applying ETAD corrections, spatial alignment improves markedly, with PS points more accurately coinciding with the true building structures. Fig. 15 further shows the vertical profile along the AA, where the corrected PS points more closely follow the rooftop elevation contours, indicating enhanced vertical positioning precision.

In Area B, encompassing a large stadium structure, was also analyzed to assess both horizontal and vertical positioning consistency. Fig. 16 presents the plan-view distribution of PS points overlaid on the LiDAR-derived DSM before and after ETAD correction. The corrected PS points exhibit markedly improved horizontal alignment with the stadium outlines. Fig. 17 shows the vertical profile along the BB, where the ETAD-corrected PS points more closely follow the actual terrain surface and stadium height, demonstrating enhanced accuracy in 3-D geolocation.

VI. Conclusion

This study proposed a PSI processing chain that systematically integrates Sentinel-1 ETAD as a standard correction stage. ETAD provides high-precision range- and azimuth-timing refinements through updated annotations of instrument delays and antenna phase center variations, together with dedicated layers for propagation-path delays (TRO and ION),

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solid-Earth tidal displacements, DOP-induced range shifts, and azimuthal biases due to FM-rate mismatch. The effectiveness of ETAD integration was evaluated across four dimensions: interferometric phase quality, deformation velocity estimation, deformation time-series accuracy, and PS geolocation.

Critically, we deliver the first fully specified, reproducible ETAD-PSI processing workflow. It is not a conceptual integration, but a production-ready pipeline that jointly refines interferometric phase and 3-D PS localization (range, azimuth, and height) within a single, physically consistent framework. This represents the first end-to-end implementation of ETAD in operational PSI—bridging the gap between ESA's calibration product and users' daily processing needs. Furthermore, our evaluation is intentionally comprehensive and directly referenceable: it quantifies impact across the full PSI value chain—from raw interferogram quality, to deformation time-series fidelity, to 3-D geolocation precision.

For interferogram quality, PSI+ETAD suppressed noise artifacts and improved fringe continuity, with the SPD reduced by 60.77% and the PSD reduced by 40.3% relative to PSI. In deformation velocity estimation, PSI+ETAD reduced the STD to 5.26 mm/year, compared with 18.64 mm/year for PSI, 2.82 mm/year for PSI+STF, and 6.34 mm/year for PSI+GACOS. These results show that PSI+ETAD provides a better balance between noise suppression and signal preservation than either PSI+STF or PSI+GACOS. For time-series analysis, PSI+ETAD lowered the SDD to 3.65 and the STC to 2.23, offering improved model conformity and effective noise mitigation. Validation against GNSS confirmed that PSI+ETAD yielded the smallest errors, with MAE of 3.18 mm and RMSE of 3.91 mm, outperforming all other variants. In terms of geolocation, PSI+ETAD introduced systematic timing refinements that produced mean offsets of 5.566 m in range, 1.736 m in azimuth, and 0.974 m in cross-range. Comparisons with airborne LiDAR validated these improvements, with corrected PS point clouds showing markedly better alignment with structural features in both planimetric and vertical dimensions.

Overall, integrating ETAD into standard PSI workflows provides a cost-effective and operationally feasible correction strategy, substantially enhancing phase stability, deformation accuracy, and geolocation precision for large-scale InSAR monitoring in urban and tectonically active environments. This work thus redefines ETAD's role: not as an auxiliary data product, but as the foundational timing calibration layer upon which robust, multidimensional PSI can be built.

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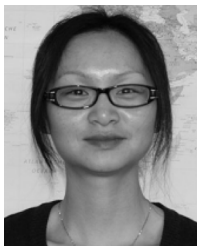
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